

1. Executive Summary

DeepMindQ is not evolving as a collection of features. It is evolving as an **intelligence maturity model**, where each phase answers a fundamentally different question about the relationship between external business events and sales opportunity. This roadmap defines three phases of intelligence evolution, each building on the previous, with clear boundaries and validation gates that prevent premature complexity.

The core product principle: sales tools provide accounts and contacts. DeepMindQ provides something radically different. It answers five questions for every intelligence item: **What changed? Why does it matter? Why are we relevant? Who should act? What should we do next?** The ultimate differentiation is not more data but better sales decisions, backed by evidence that a salesperson can trust and act on immediately.

Phase 2A proves the intelligence loop. Phase 2B proves deep understanding through knowledge graphs and entity resolution. Phase 2C proves prediction and autonomous learning. The cockpit is already built, the intelligence objects are the frozen contract, and every future engine plugs behind them without requiring redesign.

"The goal is not more data. The goal is better sales decisions. The product wins when the majority of surfaced intelligence makes a salesperson say: I did not know this, and it changes how I approach this account."

2. Phase 2A: Intelligence Loop Foundation

Current Phase

Phase 2A has a single, non-negotiable objective: prove that DeepMindQ can discover real external business changes, convert those changes into evidence-backed intelligence, and guide sales action. This is not about building the complete intelligence engine. It is about proving the loop works end to end. The validation question is: **"Did DeepMindQ find something new, explain why it matters, connect it to our capabilities, and recommend action?"** If the answer is yes, the foundation is proven.

2.1 Architecture

The Phase 2A architecture follows a linear intelligence pipeline from external information sources through evidence collection, signal detection, freshness ranking, deterministic reasoning, and finally intelligence object generation. The Intelligence Object contract remains frozen throughout. The experience layer never knows whether intelligence came from a customer upload, enrichment, news, LinkedIn, job postings, technology discovery, government sources, or AI reasoning. Everything becomes an evidence-backed Intelligence Object.

Layer	Component	Description
Collection	External Intelligence Collector	Multi-sensor: business signals, hiring, people/leadership

Layer	Component	Description
Evidence	Evidence Engine (Foundation)	Raw preservation: source, URL, date, quality, confidence
Classification	Replaceable Classifier	Rule-based keyword matching (Phase 2A); AI replacement (Phase B)
Signal	Signal Generator	CompanySignal with full Wave 8A fields
Ranking	Freshness Ranking Engine	5-dimension composite: confidence, freshness, source, business, capability
Reasoning	Deterministic Reasoning Chain	What happened, why it matters, why relevant, who acts, what to do
Output	Intelligence Objects	Frozen contract: evidence state, confidence, origin, ranking score

2.2 Three Intelligence Sensors

Phase 2A deploys three intelligence sensors. Critically, these are not features but sensors. The architecture always thinks: Company, then Multiple Sensors, then Evidence, Signals, Reasoning, and Action. Today the sensors are external intelligence, hiring, and people/leadership. Future sensors include technology intelligence, website intelligence, industry intelligence, and competitive intelligence.

2.2.1 Sensor 1: External Business Intelligence

The primary sensor detects strategic business changes through public announcements, partnerships, strategic news, and public disclosures. For enterprise companies, this is the richest signal source. When Microsoft announces an AI infrastructure expansion, that is a highly visible, easily detectable strategic signal with high confidence and clear business implications. The sensor uses targeted web search queries focused on company announcements, partnerships, acquisitions, transformation initiatives, and regulatory events.

2.2.2 Sensor 2: Hiring Intelligence

Hiring intelligence is the most important sensor for mid-market companies (200-2,000 employees). These companies may have almost zero media coverage, yet their hiring patterns reveal strategic direction. When a 500-person company hires five Azure engineers, a data architect, an AI engineer, and a cybersecurity specialist, DeepMindQ infers a possible cloud modernization initiative. This is a stronger buying signal than any news article they might publish. The sensor targets public job postings, careers pages, and hiring announcements. Phase 2A starts with individual signal detection, but the data structure allows future aggregation: ten cloud-related openings in one department means something different from one.

2.2.3 Sensor 3: People and Leadership Intelligence

The people sensor detects organizational capability changes through new appointments of CIOs, CTOs, VPs of Engineering, Digital Transformation leaders, and other senior technology roles. New technology leaders often create vendor evaluation opportunities during their first 90 days, making leadership change one of the

highest-value signals for sales engagement. The sensor uses public biographical information, indexed public information, and company announcements. It deliberately avoids LinkedIn API dependency, treating LinkedIn as a possible evidence source rather than the core engine.

2.3 Company-Size Adaptive Intelligence Strategy

DeepMindQ cannot treat Microsoft and a 300-person company identically. The intelligence strategy must change based on company maturity, using the existing sizeRange field on the Company model. Enterprise companies generate many public signals, so the system prioritizes strategic announcements, acquisitions, partnerships, investor information, and leadership changes. Mid-market companies may have almost zero media coverage, so the system prioritizes hiring patterns, leadership appointments, technology role openings, new capabilities being built, partnerships, and growth signals. Same intelligence engine, different collection strategy.

Dimension	Enterprise (10,000+)	Mid-Market (200-2,000)
Primary Signals	Announcements, acquisitions, partnerships	Hiring, technology adoption, leadership
Query Strategy	News-focused	Hiring-focused: careers + job postings + people
Signal Density	High: many public signals	Low: limited public data, hiring is primary
Confidence Model	Source quality driven	Pattern driven (aggregate signals)
Key Test	Handle information-rich companies	Find intelligence when data is sparse

2.4 Freshness Engine

The freshness engine makes intelligence behave like a living system. All signals no longer look equal. Recent intelligence wins. The engine uses a half-life exponential decay model where the decay rate is specific to each signal type. News decays fast (14-day half-life). Structural changes like regulatory events decay slowly (90-day half-life). The composite intelligence ranking score combines five dimensions: Confidence (25%), Freshness/Recency (30%), Source Quality (15%), Business Relevance (15%), and Capability Fit (15%). A fresh 85% confidence signal (3 days old, freshness approximately 82) receives a significantly higher ranking than an old 95% confidence signal (8 months old, freshness near zero).

2.5 Evidence Engine Foundation

Every intelligence item must answer the fundamental trust question: "How do you know this?" The evidence structure follows a clear chain: Claim, then Evidence (source, URL, date, quality, confidence), then Reasoning (why this matters for this company), then Recommended Action. Raw evidence is stored in the existing Evidence table with original headline, snippet, source URL, published date, collection date, and source reliability score. This becomes the Phase B Evidence Engine foundation.

2.6 Signal Taxonomy

The classifier supports ten signal types, each producing a distinct reasoning chain. Leadership change triggers "new decision-maker window." Technology adoption triggers "capability gap or transformation opportunity detected." Hiring triggers "investment pattern detected." The classification layer is isolated and replaceable: Phase 2A uses rule-based keyword matching, Phase B will replace it with an AI Evidence Engine and Knowledge Graph. The downstream Intelligence Object never changes.

Signal Type	Implication	Half-Life
funding	Available budget, active investment phase	30 days
hiring	Growth trajectory, resource gaps	21 days
leadership_change	Strategic inflection, vendor re-evaluation	45 days
expansion	Infrastructure and service needs	60 days
tech_change	Modernization, current solution dissatisfaction	30 days
partnership	Ecosystem building, integration requirements	45 days
acquisition	Technology consolidation, budget restructuring	30 days
news	Strategic priorities, market positioning	14 days
people_change	Team capability expansion	30 days
technology_adoption	Tool/platform alignment opportunity	45 days

2.7 Intelligence Origin Tracking

Every intelligence item knows where it came from. This is the foundation of enterprise trust. Origin types: customer_uploaded, enrichment, external_discovery, human_validation, ai_reasoning. Each origin includes source name and collection timestamp. A salesperson should understand: "DeepMindQ knew this because..." This transparency will become a major enterprise differentiator and is preserved across all phases.

2.8 Validation Framework

Validation is measured from a salesperson's perspective. The core question: "Can a salesperson prepare for an executive meeting in 10 minutes using DeepMindQ intelligence?" Each intelligence item is evaluated on four dimensions: Discovery (did DeepMindQ find something new?), Trust (can every recommendation be traced to evidence?), Relevance (does capability matching make commercial sense?), and Action (does the salesperson know who to contact, why now, what conversation, and what capability to position?).

2.9 Intelligence Surprise Score

Every intelligence item is rated on a 1-5 scale. Score 1: salesperson already knows this. Score 2: findable in 5 minutes. Score 3: findable eventually. Score 4: would likely miss without DeepMindQ. Score 5: genuinely surprised. The product wins when the majority scores 4 or 5. Information availability is not the value. Discovering business moments is the value.

Score	Definition	Example
1	Already known	Company has Azure on their website
2	Findable in 5 minutes	Google: "Company X hires engineer"
3	Findable eventually	Buried in careers page
4	Would likely miss	Pattern across multiple signals
5	Genuinely surprised	Cross-signal inference

3. Phase 2B: Intelligence Depth Engine

After Phase 2A proves the loop

Phase 2B moves from "detected signals" to "understood business intelligence." This is where the moat starts. The question changes from "What happened?" to "What does this mean in context?" Phase 2B adds depth through knowledge graphs, entity resolution, advanced evidence reasoning, competitive intelligence, and deeper technology discovery. It does not add new sensors; it deepens the understanding of signals already flowing through the Phase 2A pipeline.

3.1 Knowledge Graph

The knowledge graph enables deep reasoning across relationships. The system understands that Company (Microsoft), Technology (Azure), Signal (AI investment), Capability (AI modernization), People (VP AI), Case Study (similar customer success), and Recommendation (engage VP AI) are all connected. The system understands relationships, not records. Current models already preserve many of these relationships through existing tables. Phase 2B adds a formal graph layer for cross-entity traversal and inference.

3.2 Entity Resolution

The system must understand that "Microsoft Corporation," "Microsoft Azure," and "Microsoft AI" are connected entities, while "ABC Technologies Saudi" is not "ABC Technologies India." Entity resolution is critical for enterprise deployments where the CRM may contain dozens of name variants, merged entities, and subsidiary relationships. Without it, intelligence attributed to the wrong entity creates noise, destroys trust, and makes the entire pipeline unreliable.

3.3 Advanced Evidence Engine

Phase 2B moves from evidence storage to evidence understanding. Capabilities include source reliability scoring, conflicting evidence detection, multiple source confirmation, and evidence clustering. When three independent sources confirm "cloud migration underway," confidence increases. The critical test: a salesperson can defend every recommendation by pointing to specific, auditable evidence.

3.4 Competitive Intelligence

Phase 2B adds the ability to understand competitive dynamics. When a customer uses AWS, DeepMindQ detects rising Azure hiring and cloud migration signals, it identifies a possible cloud strategy shift and recommends positioning multi-cloud optimization. This requires competitor knowledge, a technology graph, and market intelligence, but requires no architectural changes because the evidence pipeline and reasoning chain already exist.

3.5 Deeper Technology Intelligence

Beyond Phase 2A's web search-based detection, Phase 2B adds technology fingerprinting through website analysis, GitHub repositories, architecture indicators, and infrastructure detection. The system builds a comprehensive technology profile across multiple channels, creating a data platform and cloud modernization opportunity map that goes far beyond what job posting keywords can reveal.

4. Phase 2C: Autonomous Intelligence Platform

Long-term vision

Phase 2C transforms DeepMindQ from a reactive intelligence system into a continuously learning sales intelligence platform. The question changes from "What happened?" to "What will happen next?" The salesperson does not search for intelligence. DeepMindQ alerts: "Three important changes happened this week. Recommended action: schedule executive conversation."

4.1 Continuous Intelligence Loop

The continuous loop operates in seven stages: Observe (monitor signals), Understand (classify and contextualize), Recommend (generate prioritized actions), Act (salesperson takes action), Measure (track outcome), Learn (update models based on results), and Improve (refine future recommendations). The system learns which recommendations worked, which messages converted, which stakeholders responded, which signals predicted opportunities, creating a flywheel effect.

4.2 Predictive Opportunity Intelligence

The system evolves from detecting current patterns to predicting future opportunities. "Companies with this hiring pattern usually purchase AI consulting within 90 days." Prediction includes opportunity probability: 82%. This requires a large dataset of historical signal-to-outcome mappings, which Phase 2A and 2B are

building the foundation for. Each signal collected, each recommendation made, and each outcome tracked feeds the prediction model.

4.3 AI Reasoning Engine

At this stage, LLM reasoning operates over validated evidence, knowledge graph, history, and outcomes. Critically, the AI never invents. It reasons over validated evidence to produce strategic analysis. The AI becomes a strategic analyst that understands context, not a content generator that produces plausible-sounding text. This distinction is the difference between a product that salespeople trust and one they ignore.

4.4 Cross-Account Intelligence

The system identifies patterns across the entire account portfolio. "Across 500 accounts, 15 manufacturing companies are showing identical AI modernization patterns. Recommendation: create industry campaign." Cross-account intelligence transforms DeepMindQ from a single-account tool into a portfolio intelligence platform, enabling sales leaders to see industry trends, regional patterns, and competitive movements.

5. Architecture Principles

5.1 The Frozen Contract

The Intelligence Object is the frozen UI/API contract. The cockpit is already built. Every future engine plugs behind the intelligence objects without requiring experience layer changes. This principle is non-negotiable. The UI should never know whether intelligence came from a customer upload, enrichment, news, LinkedIn, job postings, technology discovery, government sources, or AI reasoning. Everything becomes an evidence-backed Intelligence Object. This single architectural decision enables the entire three-phase evolution without redesign.

5.2 Source-Agnostic Collector

The external intelligence collector is an interface, not an implementation. Today it uses a search API. Tomorrow it could use APIs, crawlers, or enterprise connectors. The source can change. The intelligence pipeline should not. The current collector proves the loop. It should not become the foundation limitation. Every sensor feeds through the same evidence-signal-reasoning-output pipeline.

5.3 Replaceable Classification

The classification layer is isolated. Today: raw evidence through rule-based classifier. Tomorrow: raw evidence through AI Evidence Engine and Knowledge Graph. The downstream Intelligence Object does not know or care how classification happened. This keeps Phase B clean and prevents classification improvements from requiring changes to the reasoning chain, ranking engine, or experience layer.

5.4 Multi-Sensor Model

No single source should become a dependency. The architecture treats every source as a sensor, with the company at the center and multiple sensors (News, Website, People, Technology, Industry, Competitive) feeding into the Evidence Engine. Each source provides evidence. The system combines multiple evidence sources to produce intelligence. News is only one sensor. Hiring is only one sensor. The intelligence engine must support different intelligence patterns depending on company size and maturity.

6. Validation and Success Criteria

6.1 Phase 2A Validation Gate

Phase 2A is validated when four criteria are met across both enterprise and mid-market test companies: Discovery (DeepMindQ finds something new), Trust (every recommendation traceable to evidence), Relevance (capability matching makes commercial sense), and Action (salesperson knows who to contact, why now, what conversation). The dataset must include enterprise companies like Microsoft and 2-3 mid-market companies (200-2,000 employees, limited news coverage, active hiring pages, different industries).

Dimension	Enterprise Test	Mid-Market Test
Discovery	Rich signals: announcements, partnerships	Sparse signals: hiring, people, technology
Trust	Multiple premium sources confirm	Limited sources but clear chain
Relevance	Strong alignment with large capabilities	Targeted alignment with specific needs
Action	Clear executive narrative	Specific hiring/technology-driven actions
Surprise Score	Target ≥ 3.0	Target ≥ 4.0

6.2 Explicitly Excluded from Phase 2A

The following items are deferred to Phase 2B to maintain focus: LinkedIn API dependency (authentication limitations, commercial risk), Knowledge Graph (requires validated evidence volume), AI Reasoning Engine (requires knowledge graph), Full Technology Scanning (requires different architecture), Website Content Diff Engine (requires content snapshot storage), and Competitive Intelligence (requires technology graph). Building them prematurely would create complexity without validation.

Deferred Item	Reason for Deferral
LinkedIn API dependency	Authentication limitations, commercial risk, blocking potential

Deferred Item	Reason for Deferral
Knowledge Graph	Requires validated evidence volume from Phase 2A
AI Reasoning Engine	Requires knowledge graph and evidence depth
Full Technology Scanning	Requires website diff + deep scanning architecture
Website Content Diff Engine	Requires content snapshot storage and comparison
Competitive Intelligence	Requires technology graph and market data

7. Roadmap Summary

The DeepMindQ intelligence evolution follows three clear phases, each with a distinct question and validation gate. Phase 2A proves the system can **discover**. Phase 2B proves the system can **understand**. Phase 2C proves the system can **predict and continuously improve**. The cockpit is already built, the intelligence objects are the contract, and every future engine plugs behind them. The roadmap is frozen. Execution proceeds phase by phase.

Phase	Question	Key Components
2A	Can DeepMindQ discover?	Sensors, Freshness, Evidence, Reasoning
2B	Can DeepMindQ understand?	Knowledge Graph, Entity Resolution, AI Reasoning
2C	Can DeepMindQ predict?	Learning Loop, Prediction, Cross-Account

"Phase 2A proves DeepMindQ can discover. Phase 2B proves DeepMindQ can understand. Phase 2C proves DeepMindQ can predict and continuously improve. The ultimate differentiation: not finding information, but understanding relationships between information."